Date: __/__/20

Mth-621

Q.1

If f is differentiable at x_0, then f is continuous at x_0?

Ans:-

$$f(x) = f(x_0) + [f'(x_0) + E(x)](x - x_0)$$

where E is defined on a neighborhood of x_0 and

$$\lim_{x \to x_0} E(x) = E(x_0) = 0$$

M.M

Q.2

State the chain rule for differentiating compositions of functions?

Ans:-

~~Let~~ if g is differentiable at x_0 and f is differentiable at $g(x_0)$, then the composite function $h = f \circ g$, defined by $h(x) = f(g(x))$ is differentiable at x_0, with

$$h'(x_0) = f'(g(x_0))\, g'(x_0).$$

Note: $f \circ g$ [o is sign]

Date: __/__/20

Mth-621

Q.3

State Rolle's Theorem?

Ans:-

If a function $f(x)$ defined in a closed interval $[a,b]$ satisfying the following three conditions.

i) $f(x)$ is continuous ~~over~~ on $[a,b]$

ii) $f(x)$ is differentiable ~~over~~ on (a,b) M.M

iii) $f(a)=f(b)$

then there exist at least one point c where $c \in (a,b)$ such that $f'(c)=0$.

Q.4

Determine whether the function $f(x)=1+x$ is integrable on $[a,b]$?

Sol:-

If $f(x)=\sin x$ and $g(x)=\frac{1}{x}$, $x \neq 0$

then

$$h(x)=f(g(x))=\sin\frac{1}{x} \quad , \quad x\neq 0$$

and

$$h'(x)=f'(g(x))\,g'(x)=\left(\cos\frac{1}{x}\right)\left(-\frac{1}{x^2}\right) \quad , \quad x\neq 0$$

Mth-621

Q.5

If $(a_x)_{x \geq 1}$ is monotonically decreasing and is bounded ~~below~~ below, then prove it is convergent?

Proof:- $\therefore$ iff = if and only if

Follows from the following facts.

i) $(a_x)_{x \geq 1}$ is monotonically decreasing iff $(a_x)_{x \geq 1}$ monotonically increasing.

ii) $(a_x)_{x \geq 1}$ is bounded ~~be~~ below iff $(a_x)_{x \geq 1}$ is bounded below.

iii) $(a_x)_{x \geq 1}$ is convergent iff $(a_x)_{x \geq 1}$ is convergent.

M.M

Q.6

Define Lipschitz Continuity?

Ans:-

A function that satisfies inequality

$$|f(x) - f(x')| \leq M|x - x'|, \quad x, x' \in (a,b)$$

for all x and x' in an interval, where $M > 0$ is some real number is said to satisfy a Lipschitz condition on the interval.

Date:

Mth-621

Q-7

For the function $f(x)=\sqrt{x}$, $g(x)=\frac{9-x^2}{x+1}$ then check the continuity of $f \circ g$?

Sol:-

The function f is continuous for $x \geq 0$ and the function g is continuous for $x \neq -1$ since $g(x) \geq 0$ if $x \leq -3$ or $-1 < x \leq 3$ then the composite function

$$(f \circ g)(x) = \sqrt{\frac{9-x^2}{x+1}}$$

M.m

is a continuous on

$$(-\infty, -3) \cup (-1, 3)$$

it is also continuous from the left at -3 and 3.

Q.8

Check continuity at $x=0$, $f(x)=x+1$, $x \geq 0$
$f(x)=x-1$, $x<0$?

Ans:-

Every open interval containing $x_0=0$ also contains points x_1 and x_2 such that $|g(x_1)-g(x_2)|$ is as close to 2

Therefore,

$$\lim_{x \to x_0} g(x) \text{ does not exist.}$$

Date:

Mth621

Q.9

State change of Variable theorem?

Ans:–

Let the transformation $x=\phi(t)$ maps the interval $c\leq t\leq d$ into the interval $a\leq x\leq b$ with $\phi(c)=\alpha$ and $\phi(d)=\beta$ and let f be continuous on $[a,b]$. Let ϕ' be integrable on $[c,d]$. then

$$\int_{\alpha}^{\beta} f(x)\,dx=\int_{c}^{d} f(\phi(t))\phi'(t)\,dt$$

M.M

Q.10

Mean value theorem.

Let f be a function, then

i) $f(x)$ is continuous on $[a,b]$

ii) $f(x)$ is differentiable on $]a,b[$

then $\exists$ a point ~~c such that~~ $c\in]a,b[$

~~then~~ such for

$$f'(c)=\frac{f(b)-f(a)}{b-a}$$

CS CamScanner

Q.13 MTH-621

Date:

$f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational} \\ 1 & \text{if } x \text{ is rational} \end{cases}$

Find $\underline{\int_a^b} f(x)\,dx$

Solution

$m_j = 0$ and $M_j = 1$, $1 \leq j \leq n$

then

$$S(P) = \sum_{j=1}^{n} 1 \cdot (x_j - x_{j-1}) = b-a$$

$$s(P) = \sum_{j=1}^{n} 0 \cdot (x_j - x_{j-1}) = 0$$

Since all upper sums $= b-a$

all lower sum $= 0$

then

$$\underline{\int_a^b} f(x)\,dx = 0$$

$$\overline{\int_a^b} f(x)\,dx = b-a$$

M.M

f is continuous

Q.11 **Mth-621**

Refinement of a partition:- (3)

If P and P' are partition of $[a,b]$ then P' is a refinement of P if every partition point of P is also the partition point of P'.

M.M

Q.12

If f is integrable on $[a,b]$, then so is $|f|$ and

$$\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx \quad (5)$$

Soln:-

since $f(x) \le |f(x)|$ and $-f(x) \le |f(x)|$

where $a \le x \le b$

then we have

$$\int_a^b f(x)\,dx \le \int_a^b |f(x)|\,dx$$

and

$$-\int_a^b f(x)\,dx \le \int_a^b |f(x)|\,dx$$

hence, this is a result.

Date:

Q.14 Mth-621

$$f(x) = \begin{cases} x^3 & x \le 0 \\ x^2 \sin\frac{1}{x} & x > 0 \end{cases}$$

5

integrate one sided derivatives

Soln

We ~~first~~ calculate the one side derivative

$$f'_{+0} = \lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \frac{x^2\sin\frac{1}{x} - 0}{x-0}$$

$$= \lim_{x\to 0^+} x \sin\frac{1}{x}$$

M.M

$$= 0$$

$$f'_{-0} = \lim_{x\to 0^-} \frac{x^3 - 0}{x - 0} = \lim_{x\to 0^-} x^2 = 0$$

hence

$$f'(0) = f'_+(0) = f'_-(0) = 0$$

Q.15 Mth-621

Date:

Given an example of a function which is continuous but not Lipchitz continuous.

Ans:-

The function

$f(x) = \sqrt{|x|}$ is continuous at $x=0$

but not Lipchitz continuous at $x=0$

Q.16 Intermediate Value Theorem

if f is a function, then

$f(x)$ is continuous on $[a,b]$. M.M

$f(a) \neq f(b)$

and μ is between $f(a)$ and $f(b)$ then

$f(c) = \mu$